

An open source FORTRAN90 Code for aerodynamic design of wind turbine blades - V5 *

M. Korjahn, bewind GmbH, Rendsburg, Germany
A.P. Schaffarczyk, Kiel University of Applied Sciences, Kiel, Germany
D. Schlipf, Flensburg University of Applied Sciences, Germany
contact: Alois.Schaffarczyk@FH-Kiel.de

Version 5 (radial induction for non-planar rotors): July 31, 2026

Abstract

A specific aerodynamic design methodology has been implemented into an open source FORTRAN90 code. It is available via github: <https://github.com/Schaffarczyk/KSS/>. Version 5 adds the inclusion of the radial induced velocity for non-planar (deflected or prebent) rotors following Li et al. [15], in two variants: Madsen's engineering correction and a vortex-cylinder model. In addition a number of latent coding inconsistencies found by a systematic (LLM-assisted) static analysis have been removed.

*CFD Lab Report No. 128, Kiel, Germany

Contents

1	Introduction	6
2	The Blade-Element Momentum (BEM) formulation	6
2.1	General	6
2.2	Multiple Solutions of the BEM-Equation	7
2.3	Tip-correction	11
2.4	Glauert Correction $cT(a)$ for $a > a_c$	11
2.5	Sign convention	11
3	Code Description	11
3.1	A short note on FORTRAN	11
3.2	Numerical Approach	12
3.3	Main program	12
3.4	Memory management	12
3.5	Subroutines	12
3.6	Functions	13
3.7	Compilation	14
3.7.1	LINUX	14
3.7.2	Windows	14
3.8	Modularization	14
3.9	Work Flow	14
4	Output Files and Tools for Visualization	14
4.1	Description of Output Files	14
4.2	Description of GNUPLOT scripts	15
5	Mode: Analysis	15
5.1	Input data	15
6	Mode: Design	22
6.1	Aerodynamic Optimization	22
6.1.1	Betz' and other criteria	22
6.1.2	Change of Twist	24
6.1.3	Change of Thickness - Arrangement of Profiles	24
7	Sample cases	25
7.1	Case 1: The NREL 5MW Baseline blade	25
7.2	Case 2: A 15MW WT reference blade	28
7.3	Case 3: A 20 MW blade	30
8	Comparison with <i>Xturb</i>	30

9	V5: Radial induction for non-planar rotors	31
9.1	Motivation	31
9.2	Input: new flags in <i>Machine.in</i>	31
9.3	Model 1: Madsen correction (RadMode = 1)	31
9.4	Model 2: Vortex cylinder (RadMode = 2)	32
9.5	Results: NREL 5 MW baseline	32
9.6	Limitations and open points	33
10	V5 code revision: consistency fixes	33
11	Summary of investigated blades	34
12	Summary, Discussion and Outlook	36
13	Disclaimer	36
14	List of Abbreviations	36

Draft

List of Figures

1	Example of distributions of multiple solutions from balancing aerodynamic (lift) forces and local (axial) momentum.	8
2	Example of how a binary search algorithm finds multiple solution along a whole blade (CIG10MW)	9
3	Variation of Angle of Attack (CIG10MW)	10
4	Sample thickness distributions	18
5	Refined thickness distribution from NREL 5 MW	19
6	Thickness distribution change to decrease mass for a 15 MW reference blade [28]	20
7	Betz and Sørensen	24
8	KSS Optimisation after 5 twist loops: $a(r)$	26
9	KSS Optimisation (Own 20MW, $r_{tip} = 133m$) after 5 twist loops: some angles	27
10	cP map from KSS, including control path (magenta) and an extended pitch variation $-15^\circ < \vartheta < 20^\circ$	28
11	cP map from NREL [6]	29
12	Comparison of circulation for an own 20 MW blade	30
13	Left: radial induced velocity $u_r(r)$ from both models (NREL 5 MW, 11 m/s, 6 m tip deflection). Right: additional driving force per blade (VC model).	33

List of Tables

1	Part of a sample job description file <i>Machine.in</i> based on NREL's 5MW reference wind turbine	16
2	Sample blade description file <i>BlaDes.in</i> for NREL/IEAwind 15MW reference wind turbine	16
3	Sample thickness assignment file <i>ProThick.in</i> for NREL/IEAwind 15MW reference wind turbine	17
4	Sample Thickness distribution file <i>ThickDis.in</i> for NREL/IEAwind 15MW reference wind turbine	17
5	Sample output file <i>Bem.out</i> for NREL/IEAwind 15MW reference wind turbine	21
6	Sample aerodynamic data <i>FF21.aer</i> for NREL/IEAwind 15MW reference wind turbine	23
7	Parameters describing a design case	23
8	Rated rotor power of NREL 5MW baseline from various sources. HAWC data from Mr. Hinrichs, aerodyn engineering GmbH; bladed data from Mr. Maniok, DNV. If all values are equally weighted, $P = 5356 \pm 89$ kW results.	25
9	Comparison for NREL/IEAwind 15 MW reference wind turbine	28
10	New <i>Machine.in</i> flags in V5. Missing lines default to <i>Radial</i> = .F. and <i>RadMode</i> = 1, so old input files run unchanged.	31
11	Effect of radial induction, NREL 5 MW baseline, 11 m/s.	32
12	Summary of investigated blades	35
13	list	36

1 Introduction

Here we report on an independent FORTRAN 90/95 (F90 for short) implementation of a purely aerodynamic wind turbine blade design methodology. It is intended to be as detailed as possible although a fair amount of knowledge of wind turbine aerodynamics is necessary, see, for example, [29, 27, 26]. Some – but not all – needs from a more practical point of view (maximum chord and twist, etc.) have been taken into account. This is in contrast to [12] who tried to find out what maximum c_P may be reached without any restrictions by *practicability*.

Our basis, a lecture of Korjahn [14], presents a so-called two-point design. In a first attempt, however, we only describe an implementation of the first part: to arrive at a blade with as large a portion of the span at optimum $a(r)$ ¹ as possible. *Design point 2* consists of a certain safety against gusts which correspond to stalled parts on the blade. Therefore the AOA has to have some safety margin against stall.

Main objectives of our approach are:

- understand and formalize the approach and
- compare to other codes.

Further or additional sources from

1. Korjahn [14],
2. Bak [2],
3. Schmitz [27] and
4. Johansen et al. [12].

are recommended to complement understanding.

Important: The code is by no means fool-proof:

Bugs may still be present.

Input data and esp. the aerodynamic one has to be checked as far as possible before its use.

2 The Blade-Element Momentum (BEM) formulation

2.1 General

(Conservation of) momentum on independent annular rings [29, 21, 24] serves as the fluid mechanical foundation. Key elements of a complete BEM formulation are

¹Usually the term Betz-optimum is related to $a(r) = 1/3 = \text{const}$. Very recently [11] it has been shown that including wake-rotation and tip-loss a somewhat different distribution results: Close to the hub $a \rightarrow 0.25$ and close to the tip $a \rightarrow 0.4$.

already discussed in [7] and further refined in classical texts by Wilson et al. [24] and de Vries [21], for example. Any applied aerodynamic wind turbine investigation uses it and it is still today regarded as the *work horse*. CFD can hardly claim to be more accurate although a large number of assumptions inherent to the BEM model have been overcome by this very different approach. The most recent analysis of the BEM-method is given in [9] with some focus on mathematical rigor and optimization.

2.2 Multiple Solutions of the BEM-Equation

It was already recognized by Wilson et al. [24] p 108 ff. that under some specific circumstances the BEM-equations may not produce unique solutions. Maniaci [5] and Ning [25] further elaborate on this subject. To give an example, Fig. 1 (for a special section of a special commercial blade) shows the local thrust coefficient cT_{loc} as function of a , the axial induction, defined by $u_1 := (1 - a) \cdot U_{-\infty}$, $0 \leq a \leq 0.5$. Momentum balance gives

$$c_T = 4a \cdot (1 - a), \quad (1)$$

whereas the counter-force from lift is its proper projection into the direction of the wind. Note that this is an approximate description only, as a' (circumferential induction) is neglected as well as the tip correction (see section 2.3) and a correction of c_T for larger values of a ($a > a_c$), see section 2.4.

Our investigation for a 20 MW blade of 133 m length confirms this. It was recognized that the traditional fixed-point iteration then did not converge but simply jumped between two different values.

As a consequence we improved our implementation and search algorithms as follows:

- Firstly, scan the whole a -region $0 \leq a < 1$ in steps of 0.01. If there are solutions found by a binary search algorithm,
- start using standard fixed-point iteration until a maximum number of iterations is reached,
- switch to Newton(-Raphson) for further iterations if no convergence is reached, and finally
- use the binary search algorithm *rtbis* from [22].

Fig. 2 shows an example. Therefore the recommendation of [24], p. 31

As a guiding principle, it is suggested that the solution to be chosen in the case of multiple solutions is the one that maintains continuity of angle of attack along the blade span.

can be confirmed.

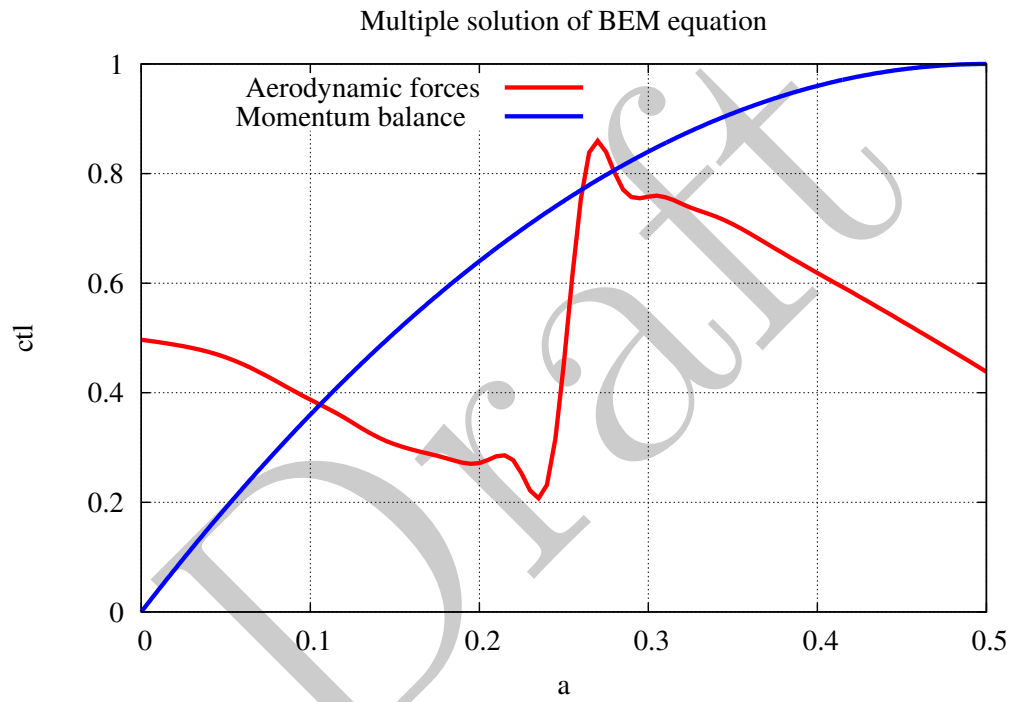


Figure 1: Example of distributions of multiple solutions from balancing aerodynamic (lift) forces and local (axial) momentum.

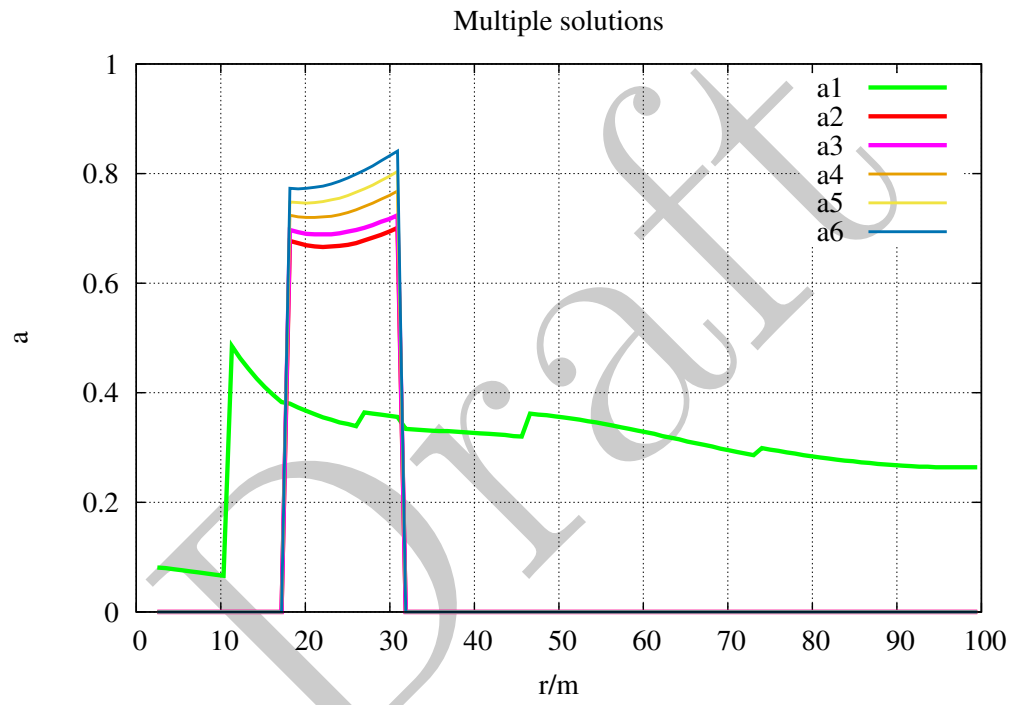


Figure 2: Example of how a binary search algorithm finds multiple solution along a whole blade (CIG10MW)

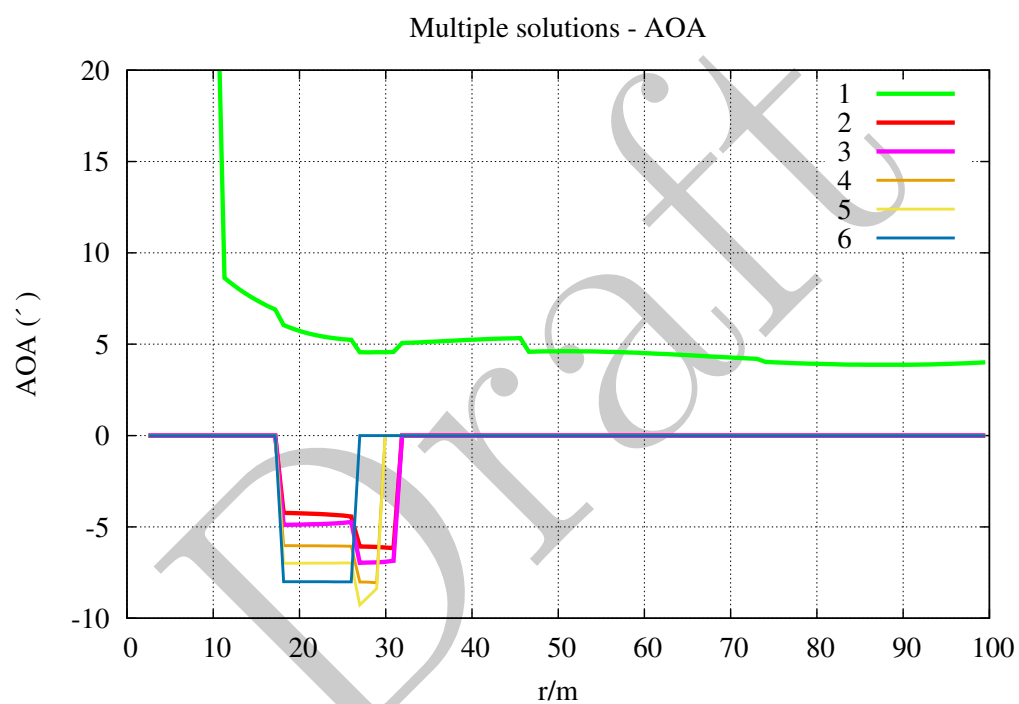


Figure 3: Variation of Angle of Attack (CIG10MW)

2.3 Tip-correction

Momentum theory is strictly valid only for an infinite number of blades, $B \rightarrow \infty$. As in most designs B is small ($B < 4$) this has to be taken into account. Indeed, as was shown very early, *tip losses* are as large as profile drag and must be accurately embedded into the code. Traditionally, the simplest (and oldest) version uses a factor F , which reduces circulation to zero when $r \rightarrow r_{tip}$:

$$F = \frac{2}{\pi} \arccos \left(\exp \left[-\frac{(B/2)(1 - r/R_{tip})}{r/R_{tip} \sin(\varphi)} \right] \right) . \quad (2)$$

Here, φ is the all-important flow angle defined by

$$\tan(\varphi) = \frac{1 - a}{\lambda(1 + a')} , \quad (3)$$

which refers to $\lambda = \omega \cdot r / U_{\infty}$ the local Tip-Speed-Ratio (TSR). $\omega = RPM \cdot \pi / 30$ is the angular velocity.

2.4 Glauert Correction $cT(a)$ for $a > a_c$

It is further well known:

- Momentum theory is valid for $0 \leq a \leq 0.5$ only, and
- for some a_c Eq. 1 becomes inaccurate.

To some extent this can be cured by *empirical engineering* extensions which are listed, for example in [9], Table 2.1. In principle at some $a_c = 0.33 \dots 0.4$ a more or less linear curve is attached to Eq. (1) so that $c_t(a = 1)$ approaches (somehow) 2 for $a \rightarrow 1$.

2.5 Sign convention

Sign issues esp. for the pitch are notorious. Following our basic interpretation

$$\alpha \text{ (AOA)} = \varphi \text{ (flow angle)} - \vartheta \text{ (twist)} , \quad (4)$$

we define **pitch** as being added to twist.

3 Code Description

3.1 A short note on FORTRAN

We will not defend our choice of the *grandfather of all (higher) programming languages* but simply use it. We have tried to write the code as plainly as possible to make understanding and changes (**i.e. maintainability**) as simple as possible. A minimum set of algorithmic features is used. In particular, no tricks from computer science have been used. This code was (and is) developed using *gfortran*, version 4.8.5 from 2015, see section 3.7. A more detailed description is given in [18].

3.2 Numerical Approach

Everybody knows from their own experience the NEWTON(-Raphson) method for finding zeros of functions ($f(x_0) = 0$) or SIMPSON's rule for 1D-Integration. The ultimate manifestation of numerical algorithms may be found in the famous (series of) text book(s): *NUMERICAL RECIPES in Fortran 77* [22]. From here we have copied

1. SPLINE and
2. SPLINT.

Unfortunately, some overshooting at sharp corners might still be possible. *Bezier* polynomials were coded independently from its definition see [19].

3.3 Main program

KSS.f contains the main code. It is merely a collection of routine calls for set-up and definition of the course of the investigation:

1. (dynamic) memory allocation,
2. Read *Machine.in*,
3. Read in spline thickness distribution from *ThickDis.in*,
4. Analyze a given set of profiles, L2D-max.
5. If design-mode is chosen:
6. An initial BEM calculation is performed and
7. chord and twist might be (Bezier-) smoothed or
8. twist may be changed to improve c_P , see section 6.1 .

3.4 Memory management

mem.f defines all GLOBAL data. Each program unit using it must start with a
use mem
statement.

3.5 Subroutines

1. BEM: BEM iteration as described in [26],
2. ANALYSIS: Analyses a blade described in *BlaDes.in* using routine *BEM*,
3. DESIGN: Determines chord and twist from given profiles and Betz [26] optimum criteria. See section 6.1 for more information about optimization.,

4. AIRFOILS: searches given α, c_L, c_D data for maximum of c_L/c_D , $c_{L,max}$ and slope at zero lift.
5. TipShapeEL: defines an elliptic (vertical tangent) tip shape geometry,
6. TipShapePara: defines an parabolic tip shape geometry determined by smooth slope at the beginning of the tip region and zero chord at tip.
7. BEZIER: smoothing of a given data set according to the definition of Bezier polynomials,
8. smoothCHORD: smooths chord with subroutine BEZIER,
9. smoothTWIST: smooths twist with subroutine BEZIER,
10. improveTWIST: (1) tries to get locally as close as possible to $1/3$ by adaption of twist, see Eq. 9. (2) J.N. Sørensen's approach, (3) a *constant circulation* approach

3.6 Functions

1. FNOVERK: (*Real Function N Over K*) calculates Binomial coefficients (BC) needed in BEZIER, (special numerical care has to be taken because of the strong growth of BCs),
2. FPR: Prandtl's tip loss F, see [26],
3. FBu: Burton's tip loss, see [23, 4],
4. chDES: optimum chord distribution from Betz (or Schmitz, or Glauert) [29],
5. chBe: Betz optimum chord, $a' \equiv 0$,
6. chSc: Schmitz optimum chord, a' is included,
7. TWIST: twist (in deg) from Betz, $a = 1/3$ and $a' = 0$, so only local TSR enters,
8. AHANSEN: gives Glauert's correction as stated in [8],
9. ANREL: gives Glauert's correction as stated in [16].

$$c_T(a) = \frac{8}{9} + \left(4F - \frac{40}{9}\right)a + \left(\frac{50}{9} - 4F\right) \cdot a^2. \quad (5)$$

10. thickpoly: cubic interpolation of thickness
11. twistbend: simple table for inclusion of twist-bend
12. apot: (1) Betz, (2) J.N. Sørensen, (3) constant circulation (very approximate)

3.7 Compilation

3.7.1 LINUX

For compilation we prefer to use (in a SUSE-LINUX environment): [18] with mandatory options (see **compDesCode.cmd**):

-fno-automatic helps to resolve notorious issues with local variables in SUBROUTINES and FUNCTIONS

-fbounds-check to avoid not allowed memory access

-O3 high optimization

A typical job with 100 sections and 100×100 wind-speed and pitch variation takes a few minutes on our simple *Intel(R) Core(TM) i3-4160 CPU @ 3.60GHz* CPU.

3.7.2 Windows

MinGW (= minimal gcc for Windows) was used to compile a Windows executable file, which is also available.

3.8 Modularization

Apart from mem and KSS the following files

- mem.f, declares global variables,
- Sub.f, a set of used subroutines and functions,
- SubNum.f, numerical functions: SPLINE and SPLINT from [22],

help to put some structure into the code.

3.9 Work Flow

All input has to be provided as ASCII files, and all output is written in the same format. Post-processing then has to be performed by other tools. We prefer GNUPLOT [20, 10] see section 4.2.

4 Output Files and Tools for Visualization

4.1 Description of Output Files

- *Bem.out*: sectional output from BEM, see table 5,
- *des.out*: sectional output from design: rsec, twist, chord, thick, c_L^{des} and AOA_{des} ,
- *ProProp.out*: from routine AIRFOILS: design properties of given airfoil data.

4.2 Description of GNUPLOT scripts

- *a.gpl*: plots $a(r)$. Data from *Bem.out*
- *chord.gpl*: plots $c(r)$. Data from *Bem.out*, *BlaDes.in* and *des.out*
- *cL-aoa.gpl* and *cL-cD.gpl*: plots polars. Data from **.aer*
- *cL-des.gpl* plots $c_L^{des}(r)$. Data from *PrOpt/Bem.out*
- *twist.gpl*: plots $\vartheta(r)$. Data from *Bem.out*, *BlaDes.in* and *des.out*
- and some more

5 Mode: Analysis

5.1 Input data

The code is able to operate in two modes:

1. Design mode, set bit *DesMode* to .T. in file *Machine.in*,
2. if not, then the code will operate to analyze a given blade.

Then, in addition, other files:

- *BlaDes.in* (r , twist, chord, attached profile name),
- *ProThick.in* which maps airfoil names to their (relative) thickness,
Important: This file has to have a header line
- *ThickDis.in* which gives with few entries r/R_{tip} and corresponding t/c ,
- **.aer*: files with airfoil data $AOA(\alpha)$, c_L , c_D , c_M (not used).

are needed. Most of the input should be more or less self-explanatory. Lines with 3 numbers are: first value, last value and number of values (wind speed and pitch).

Examples are given below.

Fig. (5) and Fig. (4) give an impression of what kind of thickness distributions have been and will be used.

Recently, [28] investigated a 15 MW reference blade in more detail. For reducing the mass from 71 t to 63 t a considerable change of thickness (see Fig. 6) was necessary. This underlines again the importance of this file.

Main output file is *Bem.out*

```

DesMode      .F.
BladeNo      3.0
dens         1.225
Rhub         0.3
Rtip         63.0
windrage     11.4 11.4 1
rpm          12.1
Pitchr       0.0 0.0 1
Nsec         70
. . . . .

```

Table 1: Part of a sample job description file *Machine.in* based on NREL’s 5MW reference wind turbine

```

r      Twist    chord    Profile Name
0.0000 15.5946  5.2000  CYL1
2.4000 15.5911  5.2083  CYL2
4.8000 15.4288  5.2357  FF50
7.2000 14.9891  5.2889  FF50
9.6000 14.3263  5.3606  FF50
12.0000 13.4947  5.4433  FF50
15.0000 12.3001  5.5510  FF50
18.0000 11.0323  5.6501  FF50
20.2841 10.0834  5.7110  FF36
. . . . .
118.7179 -1.3914  1.7239  FF21
118.9744 -1.3628  1.6717  FF21
119.2308 -1.3336  1.5813  FF21
119.4872 -1.3038  1.4157  FF21
119.7436 -1.2734  1.1027  FF21
120.0000 -1.2424  0.5000  FF21

```

Table 2: Sample blade description file *BlaDes.in* for NREL/IEAwind 15MW reference wind turbine

NAME	rel thickness
CYL1	1.0
CYL2	0.95
FF50	0.5
FF36	0.36
FF33	0.33
FF30	0.3
FF27	0.27
FF24	0.24
FF21	0.21

Table 3: Sample thickness assignment file *ProThick.in* for NREL/IEAwind 15MW reference wind turbine

0.0000	1.0
0.0200	0.98
0.1500	0.500
0.2452	0.360
0.3288	0.330
0.4392	0.301
0.5377	0.270
0.6382	0.241
0.7717	0.211
1.0000	0.211

Table 4: Sample Thickness distribution file *ThickDis.in* for NREL/IEAwind 15MW reference wind turbine

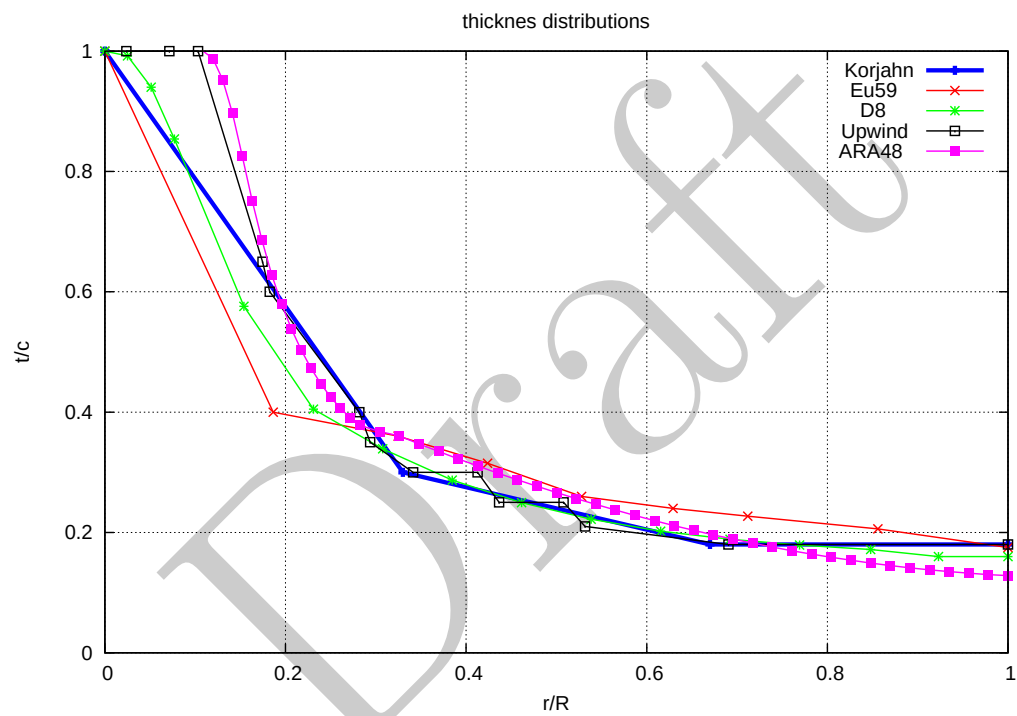


Figure 4: Sample thickness distributions

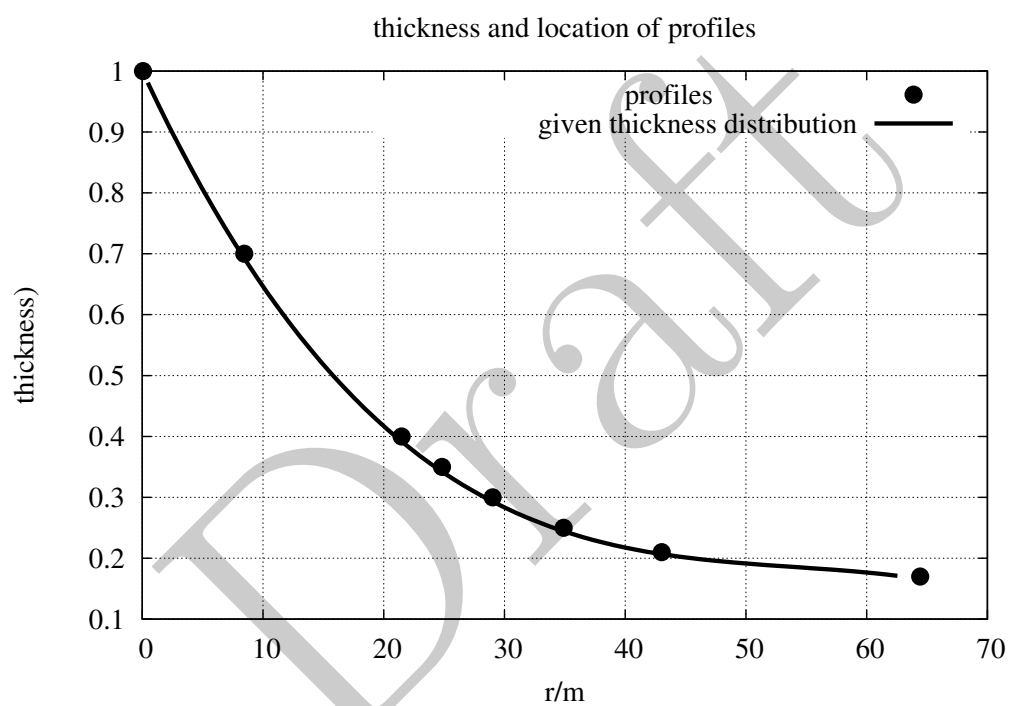


Figure 5: Refined thickness distribution from NREL 5 MW

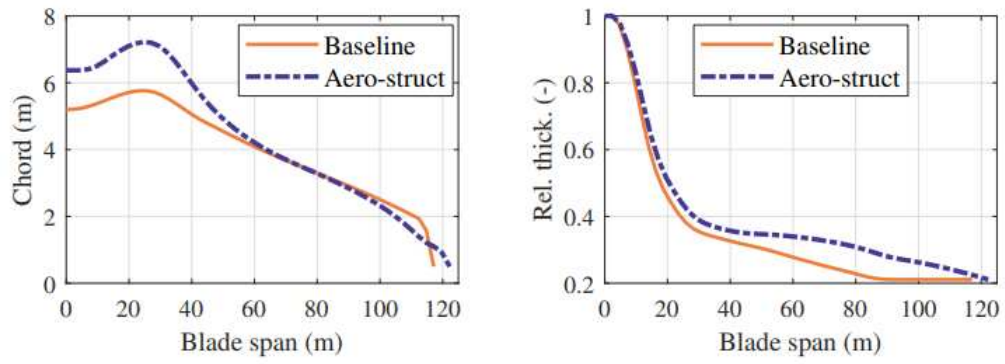


Figure 4: Chord and relative thickness distributions.

Figure 6: Thickness distribution change to decrease mass for a 15 MW reference blade [28]

r	a	aP	F	w	chord	twist	phi	aoa	cL	cD	L2D	cNo	cTa	dT	dQ	iter	err	th	Prof
0.598	0.334	-0.334	1.0	6.004	5.202	15.596	87.4	71.8	0.000	0.481	0.000	0.481	-0.022	0.20	-0.01	3	0.0E+00	0.99	CYL2
1.794	0.134	-0.134	1.0	7.861	5.205	15.598	82.3	66.7	0.000	0.444	0.000	0.440	-0.059	0.31	-0.04	3	0.0E+00	0.98	CYL2
2.990	0.076	-0.076	1.0	8.519	5.213	15.573	77.3	61.7	0.000	0.388	0.000	0.379	-0.085	0.31	-0.07	3	-0.2E-06	0.97	CYL2
4.186	0.045	-0.045	1.0	9.009	5.226	15.495	72.5	57.0	0.000	0.303	0.000	0.289	-0.091	0.27	-0.08	3	-0.2E-06	0.95	CYL2
5.382	0.040	-0.020	1.0	9.353	5.246	15.347	67.5	52.2	0.060	0.303	0.197	0.303	-0.061	0.31	-0.06	4	-0.1E-04	0.92	FF50
6.578	0.038	-0.002	1.0	9.730	5.273	15.128	62.8	47.7	0.144	0.297	0.484	0.330	-0.008	0.36	-0.01	4	-0.3E-04	0.89	FF50
7.774	0.040	0.011	1.0	10.149	5.305	14.848	58.4	43.5	0.245	0.283	0.864	0.370	0.060	0.44	0.07	5	-0.9E-07	0.85	FF50
8.970	0.043	0.021	1.0	10.608	5.340	14.519	54.2	39.7	0.356	0.263	1.354	0.421	0.135	0.56	0.18	6	0.0E+00	0.81	FF50
10.16	0.049	0.028	1.0	11.106	5.379	14.144	50.4	36.3	0.469	0.238	1.966	0.482	0.210	0.70	0.31	6	0.0E+00	0.76	FF50

Table 5: Sample output file *Bem.out* for NREL/IEA wind 15MW reference wind turbine

6 Mode: Design

If the design flag is switched on, the lower part of *Machine.in* is important and will be used. As the Betz and/or Schmitz chord distributions usually give non-zero chord at all sections, a tip length must be given within which the chord goes down to zero. Two shapes can be chosen:

- Parabolic "P" and
- Elliptic "E".

The latter gives a vertical tangent at the tip and the former is a simple quadratic parabola defined by two points (begin and end (=rtip)) and slope of $c(r)$ at the begin of the tip region.

Note: *ImpChord* flag is not used at the moment. Improvement of chord (to proceed to more monotonic-like behavior, see [14] sheet 11) is achieved by smoothing with *Bezier* polynomials.

In addition to Korjahn's $a = 1/3$ approach *DesSchema* = 1, we use *DesSchema* = (2,3) for other Schemas (with very low differences), as described in section 6.1.

6.1 Aerodynamic Optimization

6.1.1 Betz' and other criteria

We will not give a complete list or discuss in much detail the various interpretations of what an optimization might be and how this may be reached. Only the following three are implemented:

- Betz: $a(r) = 1/3 = \text{const.}$,
- Sørensen's (2022), modified *Glauert's* approach[11], see Fig. 7 and
- Joukowski's: $\Gamma(r) = \text{const.}$

(1) Betz: no further explanation needed.

(2) Sørensen [11] argued that close to the tip $a \rightarrow 2/5$, see Fig. 7.

(3) The last one is more subtle; to first order (in a) it is equivalent to Betz', but to 2nd order:

$$a(1 - a) = \frac{\lambda B \Gamma (1 + a')}{4\pi}. \quad (6)$$

To complete, a model for a' has to be added, for example

$$a' = \frac{1 - 3a}{4a - 1} \text{ (Glauert) or} \quad (7)$$

$$a' = \frac{2}{9} \frac{1}{\lambda_r^2} \text{ (Simple approximation) .} \quad (8)$$

DesSchema = 1 refers to Betz' aerodynamic optimized blade $a = 1/3$ for all spans.

.				
10.00000	-0.88209	0.01386	-0.04397	
-8.00000	-0.62981	0.01075	-0.05756	
-6.00000	-0.37670	0.00882	-0.06747	
-4.00000	-0.12177	0.00702	-0.07680	
-2.00000	0.12810	0.00663	-0.08283	
-1.00000	0.25192	0.00664	-0.08534	
0.00000	0.37535	0.00670	-0.08777	
1.00000	0.49828	0.00681	-0.09011	
2.00000	0.62052	0.00698	-0.09234	
3.00000	0.74200	0.00720	-0.09447	
4.00000	0.86238	0.00751	-0.09646	
5.00000	0.98114	0.00796	-0.09828	
6.00000	1.09662	0.00872	-0.09977	
7.00000	1.20904	0.00968	-0.10095	
8.00000	1.31680	0.01097	-0.10163	
9.00000	1.42209	0.01227	-0.10207	
.				

Table 6: Sample aerodynamic data *FF21.aer* for NREL/IEAwind 15MW reference wind turbine

.	
Tiplen	3.0
twmax	20.
chmax	5.5
ImpChord	.F.
ImpTwist	.F. 1
ImpThick	.F.
TwistB	.F.
DesSchema	2

Table 7: Parameters describing a design case

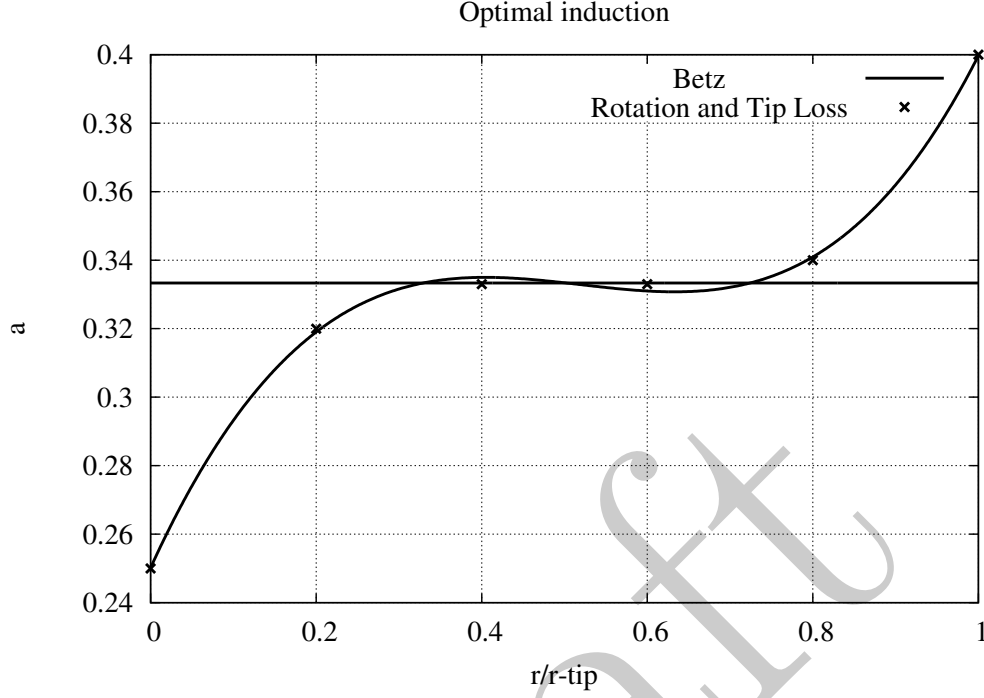


Figure 7: Betz and Sørensen

DesSchema = 2 refers to Sørensen's different $a(r)$ distribution, [11]. Fig. 7. See [12] for further discussion about aerodynamic optimization and the textbooks in the list of references, esp. [27].

DesSchema = 3 is implemented but seems not to give any significant changes. In any case we will only change twist for optimization.

6.1.2 Change of Twist

Our approach to relate twist and a runs via the flow angle and reads as: If $da := a_{opt}(r) - a_{loc} \neq 0$ then locally change twist by

$$d\varphi = -\frac{1}{\lambda_r(1 + a')}da . \quad (9)$$

6.1.3 Change of Thickness - Arrangement of Profiles

As stated in [14] (sheet 19, first figure) a further means to improve performance is to reduce thickness in the inner part. This is provided in our code by setting the *ImpThick*-bit and providing a changed thickness distribution *ThickDis.new*.

At the very end of *Machine.in* a flag *TwistB* (binary) can be set. It is a very simple way to take *pre-twist* into account. If set to .T., a table of additional twist deformation (along the blade) as a function of wind speed has to be given.

7 Sample cases

7.1 Case 1: The NREL 5MW Baseline blade

As this turbine [13] was also used by Korjahn [14] as a sample case, we first report on output from KSS as well.

Table 8: Rated rotor power of NREL 5MW baseline from various sources. HAWC data from Mr. Hinrichs, aerodyn engineering GmbH; bladed data from Mr. Maniok, DNV. If all values are equally weighted, $P = 5356 \pm 89$ kW results.

Source	Value/kW	Deviation (%)
TP 38060	5267	0.0
KSS	5248	-0.4
FAST	5293	0.5
bladed	5322	1.0
HAWC2	5323	1.1
q-blade	5364	1.8
Xturb	5418	2.9
WZX	5478	4.0
wt-perf	5494	4.3

Except for the (outdated) *wt-perf* (and some other codes) accuracy is within a 1 % range. The difference of KSS to TP38060 is only 0.4 %. This may serve as a hint that this BEM implementation is not totally wrong.

To have a glimpse of the *design capability*, Fig. 8 shows $a(r)$ after 5 loops of twist optimization. cP improves from 0.4682 to 0.4685. We agree with [14], sheet 16, that

For radius < 15 m optimum induction is not possible.

In addition, Fig. 9 shows some angles for further assessment.

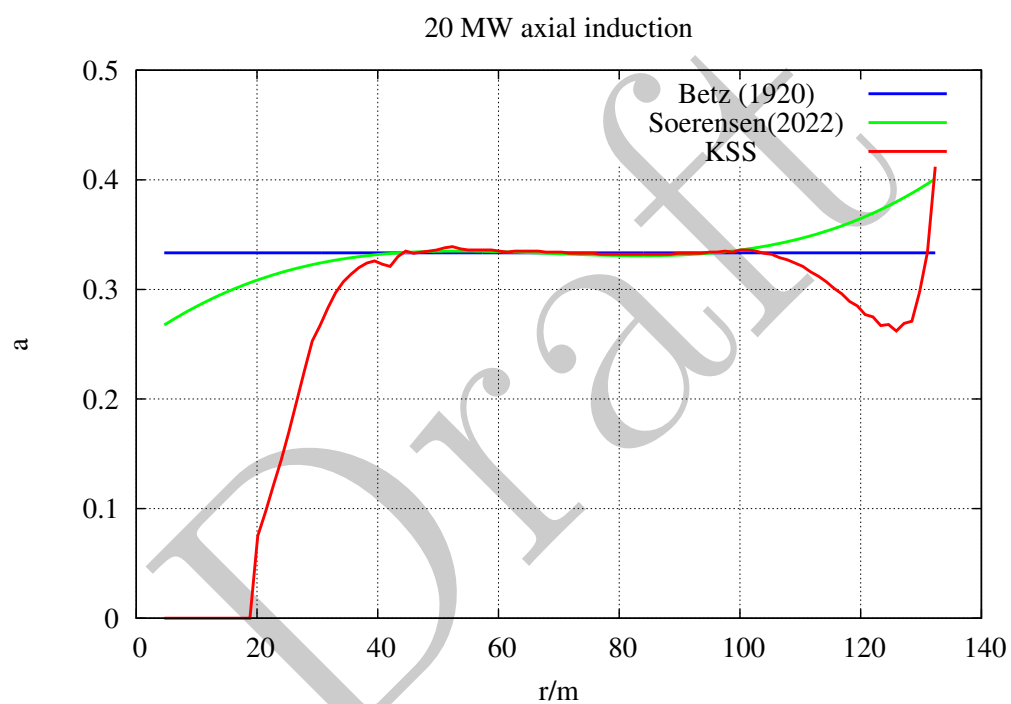


Figure 8: KSS Optimisation after 5 twist loops: $a(r)$

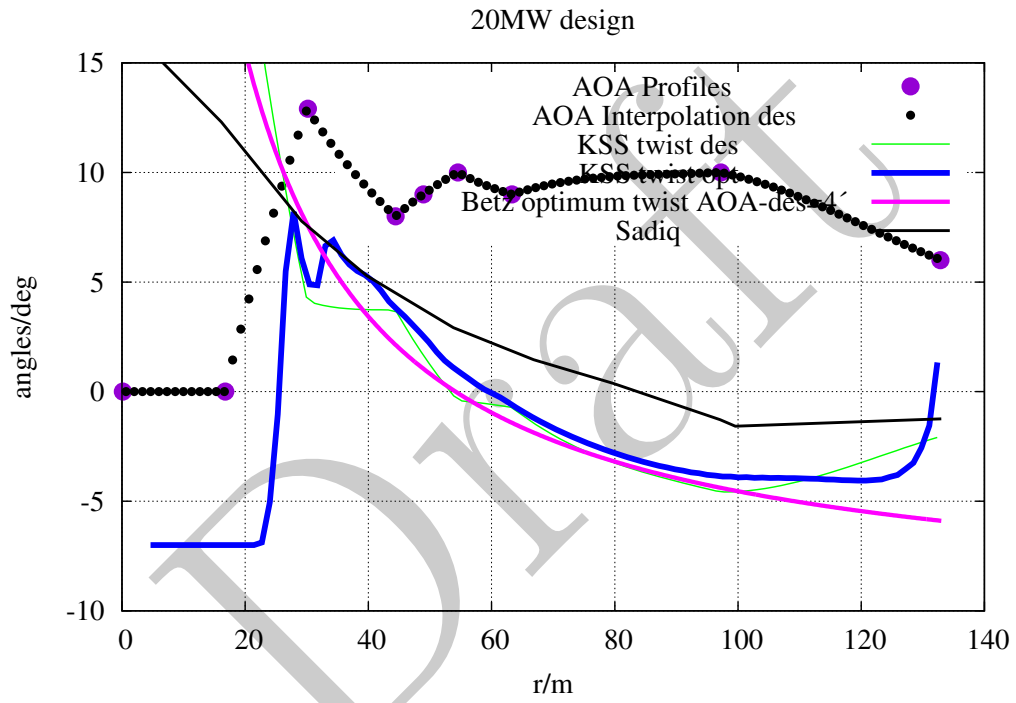


Figure 9: KSS Optimisation (Own 20MW, $r_{tip} = 133m$) after 5 twist loops: some angles

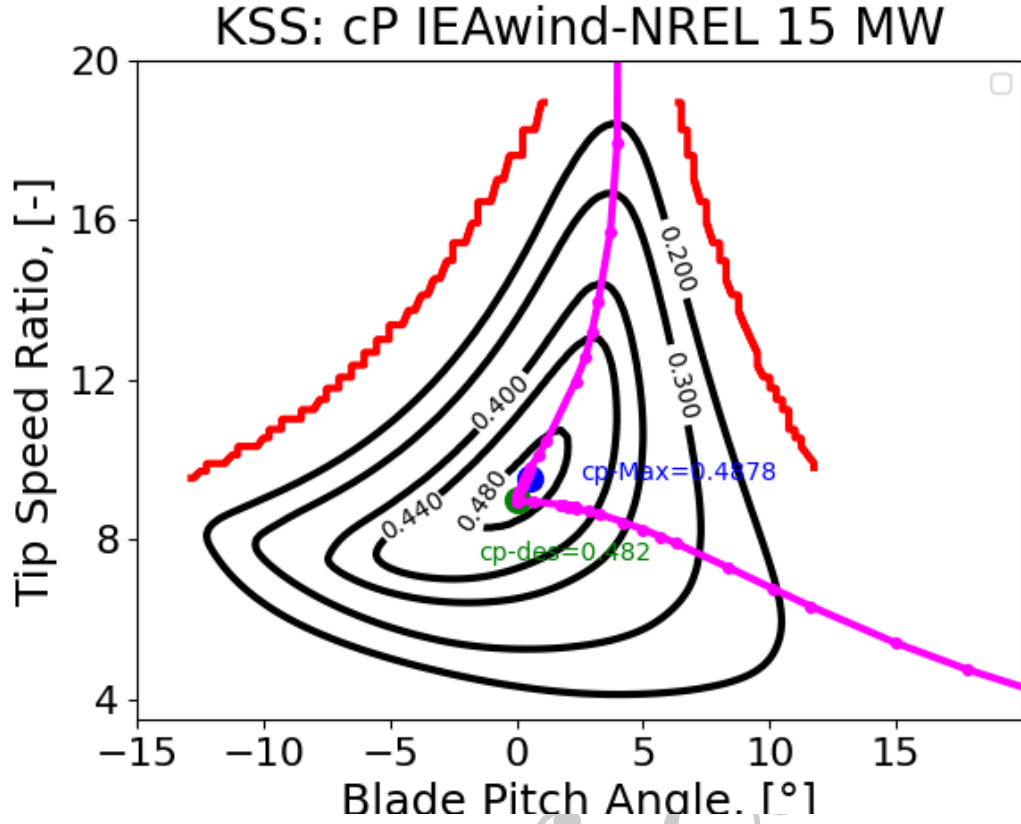


Figure 10: c_P map from KSS, including control path (magenta) and an extended pitch variation $-15^\circ < \vartheta < 20^\circ$.

7.2 Case 2: A 15MW WT reference blade

The blade described in [6] was also analyzed with KSS, see Figs. 10 and 11. Some differences are visible, but agreement at rated conditions, see table 9 (comparing c_P and c_T), is again within the 1% range.

Table 9: Comparison for NREL/IEAwind 15 MW reference wind turbine

Source	c_P	c_T
TP 76698	0.489	0.804
KSS	0.484	0.768

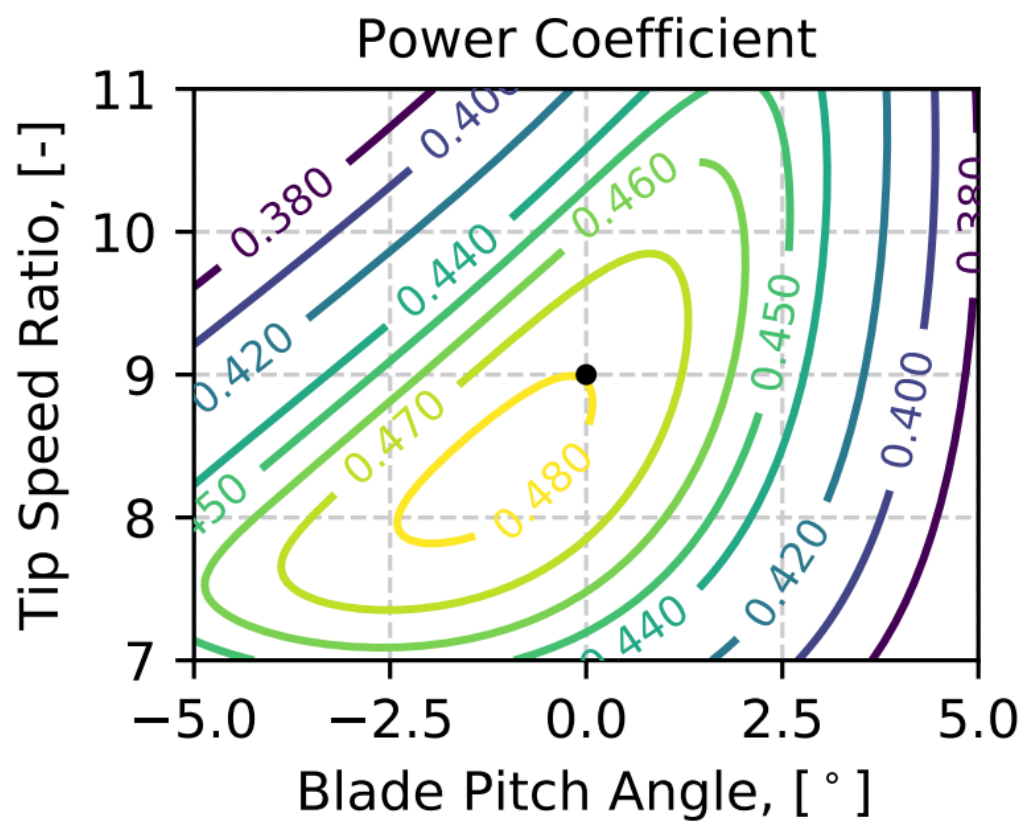


Figure 11: cP map from NREL [6]

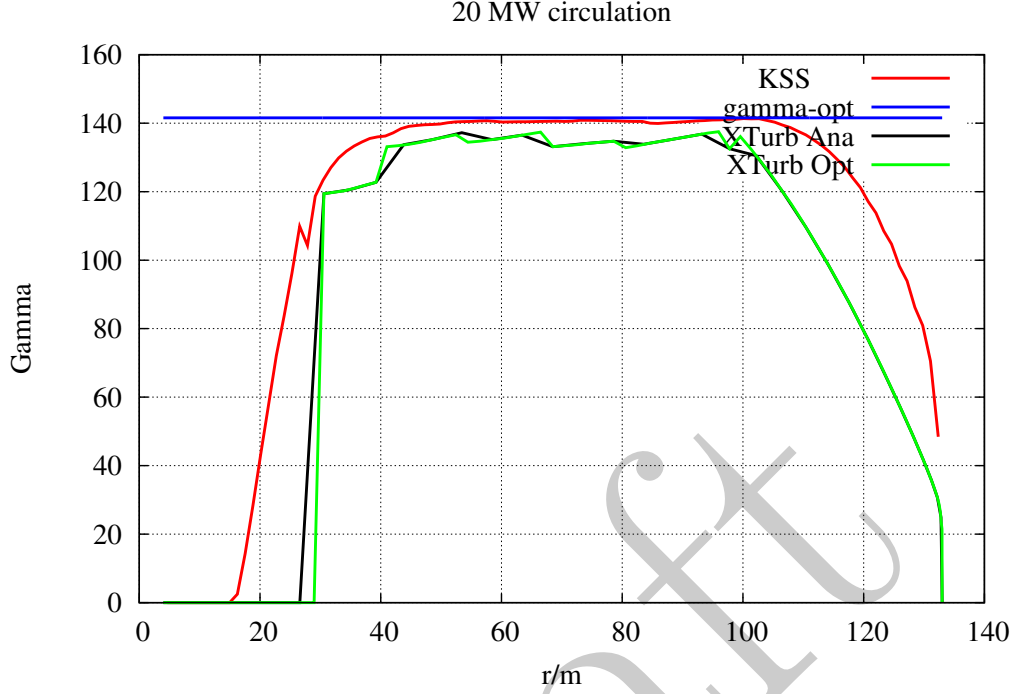


Figure 12: Comparison of circulation for an own 20 MW blade

7.3 Case 3: A 20 MW blade

[30] investigated the aerodynamic design of a 20MW blade in more detail. A maximum $cP = 0.4826$ was reached by Betz-mode of optimization after 6 loops and an initial design with $cP = 0.4821$. This is less than 1%. It seems that the restrictions are so strong that noticeable improvement is difficult.

8 Comparison with *Xturb*

Xturb by Sven Schmitz [27] offers some tools for optimization as well. We compared our 20 MW design [30], see Fig. 12. Unfortunately, XTurb showed an unexpected and non-explainable decrease close to tip ($r/R_{tip} > 0.77$) which makes a reasonable comparison impossible. Nevertheless, the initial design can be improved from $cP_{xT,A} = 0.4299$ to $cP_{xT,D} = 0.4347$ which is about 1.1% higher.

9 V5: Radial induction for non-planar rotors

9.1 Motivation

Classical BEM treats the rotor as planar and retains only the axial and circumferential induced velocities. For blades with substantial out-of-plane (oop) deflection or prebend — i.e. all recent multi-MW designs — the radial induced velocity u_r acts on the locally tilted blade section and produces an additional in-plane (driving) force. Li et al. [15] give a computationally efficient framework for this effect; see also Madsen et al. [17] and the vortex cylinder model of Branlard and Gaunaa [3]. Two of these model variants have been implemented in KSS (V5) within the scope of the MSc thesis project of B. Masare.

9.2 Input: new flags in *Machine.in*

Two lines were appended to *Machine.in*:

```
Radial      .T.      main switch (.F. = planar BEM)
RadMode     2        1 = Madsen correction (Eqs. 6+7 of Li et al.)
                  2 = vortex cylinder (Sec. 3
of Li et al.)
```

Table 10: New *Machine.in* flags in V5. Missing lines default to *Radial* = .F. and *RadMode* = 1, so old input files run unchanged.

The local cone (tilt) angle $\kappa = \arctan(dz_{oop}/dr)$ is computed from a fifth column (oop-deflection in m) appended to *Blades.in*; $\cos \kappa$ enters flow angle and relative velocity. Positive oop-deflection (downwind) gives positive κ and a positive driving-force increment; note that in [15] κ is defined positive when tilting *upwind*.

9.3 Model 1: Madsen correction (RadMode = 1)

The area-averaged thrust coefficient up to radius r ,

$$c_{T,av}(r) = \frac{\int_0^r c_T(r^*) 2\pi r^* dr^*}{\pi r^2}, \quad (10)$$

drives the radial induced velocity (exact logarithmic form of Eq. (6) in [15]),

$$u_r(x) = \frac{U_0}{2.24} \cdot \frac{c_{T,av}}{4\pi} \ln \left[\frac{0.04^2 + (x+1)^2}{0.04^2 + (x-1)^2} \right], \quad x = r/R_{tip}. \quad (11)$$

The additional tangential (driving) force per blade of the tilted section is obtained via Kutta–Joukowski from the bound circulation $\Gamma = \frac{1}{2} W c c_L$:

$$dF_{z,inc} = \rho \Gamma u_r \tan(\kappa) dr. \quad (12)$$

The correction is applied once per section *after* convergence of the induction iteration (one-way coupling, no feedback into the *a*-iteration).

9.4 Model 2: Vortex cylinder (RadMode = 2)

The converged axial induction factors a_i of the BEM sweep define $N_{sec} + 1$ semi-infinite vortex cylinders of strength

$$\gamma_t^{(k)} = 2U_0 (a_k - a_{k-1}), \quad a_0 = a_{N+1} = 0, \quad (13)$$

located at the annulus boundaries R_k and starting at the axial position of the *deformed* blade (non-planar rotor, Sec. 3.3 of [15]). The radial velocity induced by one cylinder at evaluation point (r, x) is

$$u_r(r, x) = -\frac{\gamma_t}{2\pi} \sqrt{\frac{R_k}{r}} \left[\frac{2 - k^2}{k} K(k^2) - \frac{2}{k} E(k^2) \right], \quad k^2 = \frac{4rR_k}{(R_k + r)^2 + x^2}, \quad (14)$$

with complete elliptic integrals K and E (polynomial approximation [1], 17.3.34–37, error $< 2 \cdot 10^{-8}$). u_r at a blade section is the superposition of all cylinders; the driving force increment again follows from Eq. (12). Sectional details are written to a new output file *VC.out* ($r, a, \kappa, u_r^{VC}, dF_{z,inc}$).

9.5 Results: NREL 5 MW baseline

Table 11 compares the three modes for the NREL 5 MW baseline ($v = 11$ m/s, 12.1 rpm, pitch 0° , $N_{sec} = 70$) with an assumed operational oop tip deflection of 6 m ($\kappa \approx 5.7^\circ$). The additional VC driving moment amounts to +61.8 kNm, i.e.

Table 11: Effect of radial induction, NREL 5 MW baseline, 11 m/s.

	Radial .F. (planar)	RadMode 1 (Madsen)	RadMode 2 (vortex cyl.)
Thrust / kN	699.77	695.12	695.12
Power / kW	4718.40	4726.30	4728.72
c_P	0.4642	0.4649	0.4652
c_T	0.7572	0.7522	0.7522

$\Delta c_P / c_P \approx +0.2\%$ at this small κ — consistent with the $\tan \kappa$ -scaling of Eq. (12).

Fig. 13 compares the two u_r models.

The close agreement of the two models in the *magnitude* of u_r over the outer half of the blade (within 10–35 % for $0.7 < r/R < 0.92$) is *not* a coincidence: Madsen’s engineering expression, Eq. (11), is itself a calibrated fit to the radial velocity field of the vortex-cylinder (uniformly loaded actuator disc) model, so wherever the local loading resembles the area average, both must produce the same u_r . They deviate exactly where their respective assumptions break: (i) in the mid-span region, where the discrete VC superposition responds to the non-monotonic $a(r)$ ($\gamma_t \propto \Delta a$) and u_r even changes sign, while the Madsen fit, driven by the smooth $c_{T,av}$, cannot; (ii) at the very tip, where the discrete VC sum diverges logarithmically ($u_r = 6.6$ vs. 2.3

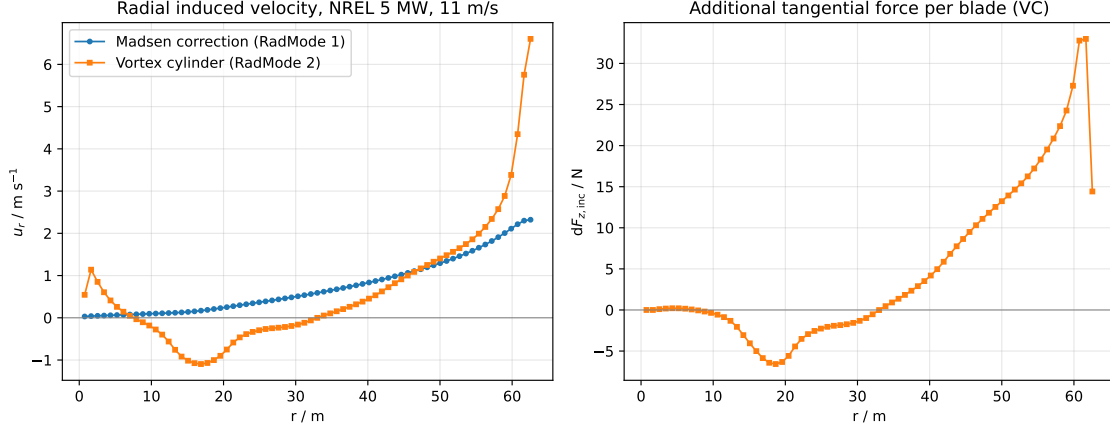


Figure 13: Left: radial induced velocity $u_r(r)$ from both models (NREL 5 MW, 11 m/s, 6 m tip deflection). Right: additional driving force per blade (VC model).

m/s at the outermost section) — an N_{sec} -dependent artefact. Since $\Gamma \rightarrow 0$ at the tip and the mid-span contributions are small and partially cancelling, the Γ -weighted integral — the additional driving moment — is nevertheless almost identical for both models (4726 vs. 4729 kW). Identical thrust is trivial: both corrections act after the induction iteration and do not alter the axial momentum balance.

9.6 Limitations and open points

- **One-way coupling:** cylinder strengths are formed from the *converged* BEM inductions; the fully coupled iteration of [15], Sec. 3.2, is a possible next step.
- **Tip cylinder:** u_r diverges logarithmically at the tip cylinder; values of the outermost sections therefore depend on N_{sec} . A systematic N_{sec} convergence study remains to be done.
- **Sign changes:** in the mid-span region u_r^{VC} may become negative. This follows from $\gamma_t \propto \Delta a$ and the non-monotonic $a(r)$ of the BEM solution and is not a coding error.
- Only u_r enters (via Eq. 12); the axial VC induction does *not* replace the momentum balance of the BEM.

10 V5 code revision: consistency fixes

In preparation of V5 the complete source was checked statically (LLM-assisted review, 2026 July). The following latent inconsistencies — most of them uninitialized variables that stayed unnoticed due to *-fno-automatic* zero-initialization — were removed:

1. *Sub1.f*: u_r , $c_{L,0}$, $c_{D,0}$ were used *undefined* in the first pass of the former in-loop radial correction; block replaced by the post-convergence correction of section 9.
2. *rtbis*: interpolation weight was never set (new module variable).
3. *ctsec*: a_c and x (tip loss argument) undefined; airfoil interpolation used the same polar twice with interchanged weights.
4. Duplicated output and limiter blocks removed.
5. *Airfoils*: undefined output unit number; index of the zero-lift search could stay undefined for cylinder profiles.
6. *fnoverk*: the fixed-form line `if (2*k.gt.n) then k=n-k` was parsed as an assignment to an implicit variable `thenk` (!) — i.e. the binomial symmetry shortcut was silently never active; corrected using a local copy of the dummy argument.

None of these fixes changed the reference results of Table 8 beyond the last printed digit, but they remove undefined behaviour.

11 Summary of investigated blades

Table 12 shows a summary of investigated blades so far:

Table 12: Summary of investigated blades

Name	Rated Power	Rotor Diameter/m
SkyWind NG	0.6 - 1 kW	1.5
Baltic Thunder	$\sim 1 - 5$ kW	1
Braak	~ 10 kW	101
SWT	1 kW	1.5
SWT(WINDFLOH)	14 kW	10.3
NASA-AMES	15 kW	11
E30	150 kW	30
GROWIAN	1000 kW	100
NecMicon 80	2 MW	80
MM92	2 MW	92
NREL-Baseline	5 MW	126
DTU	10 MW	178
CIG10MW	10 MW	200
		220
NREL	15 MW	239
	17 MW	248
KUAS20	20 MW	284
22-RWT	22 MW	266

12 Summary, Discussion and Outlook

Based on the lecture notes from [14] an F90 blade analysis and design tool has been implemented. It was applied to and tested on a wide range of WT blades, see table 12. Agreement with other sources/codes within a 1% range or better seems possible.

Findings:

- Choice of thickness distribution and polar data has probably the strongest influence on performance data for a wind turbine blade.
- Improvement (of c_P) by the proposed 1-point algorithm seems to be within a 1%-range only.
- Inclusion of radial induction (V5) increases c_P by about 0.2 % for the NREL 5 MW baseline at 6 m tip deflection; the two implemented models agree in the integral effect but differ locally at mid-span and tip (see section 9).

13 Disclaimer

As mentioned earlier, a set of **meaningful** input data has to be provided. There is almost no **consistency** check of input data and this implies that the user has to have some knowledge to avoid crashing of the code and/or meaningless output. Again and again we emphasize:

Theorem 1. *You can't calculate what you haven't understood.*

Theorem 2. *Only start the calculation, if you know the answer.*

14 List of Abbreviations

Table 13: list

	meaning
CIG	Chinese, India, Germany
DNV	Det Norske Veritas
DTU	(The) Danish Technical University
HAWC	Horizontal Axis Wind turbine simulation Code
KUAS	Kiel University of Applied Sciences
NREL	National Renewable Energy Laboratory (USA)
SWT	small wind turbine

Acknowledgments

The V5 radial-induction extension was prepared within the MSc thesis project of Bhimarao Masare (Kiel University of Applied Sciences, 2026). The static consistency review and the reference implementation of Eqs. (10)–(14) were carried out with LLM assistance (Anthropic Claude, 2026).

Draft

References

- [1] M. Abramowitz and I. A. Stegun. *Handbook of Mathematical Functions*. Dover, New York, 1964.
- [2] C. Bak. Aerodynamic design of wind turbine rotors. page chapter 3, 2013.
- [3] E. Branlard and M. Gaunaa. Cylindrical vortex wake model: right cylinder. *Wind Energy*, 18:1973–1987, 2015.
- [4] T. Burton, N. Jenkins, E. Bossanyi, D. Sharpe, and M. Graham. *Wind Energy Handbook, 3rd. Ed.* John Wiley and Sons, Hoboken, NJ, USA, 2020.
- [5] D.C Maniaci. An Investigation of WT-Perf Convergence Issues. In *49th AIAA Aerospace Sciences meeting*.
- [6] Gaertner, E. and Rinker, J. and Sethuraman L and Zahle, F. and Anderson, B. and Barter, G. and Abbas, N. and Meng, F. and Bortolotti, P. and Skrzypinski, W. and Scvott, G. and Feil, R. and Bredmose H. and Dykes, K. and Shileds, M and Alen, Chr. and Vislli, A. Definition of the IEA 15-Megawatt Offshore Reference Wind Turbine. NREL/TP-5000-75698, NREL, March 2020.
- [7] H. Glauert, editor. *Airplane Propellers*. Reprint: Stanford University, CA, USA, 1935.
- [8] M. Hansen. *Aerodynamics of Wind Turbines, 3rd. Ed.* Routledge, London, UK, 2015.
- [9] J. Ledoux and S. Rizzo and J. Salomon. Analysis of the Blade Element Momentum Theory. *SIAM Journal on Applied Mathematics*, (81(6)):2596–2621, 2021.
- [10] P. K. Jannert. *gnuplot IN ACTION, 2nd Ed.* Manning Publications, 2016.
- [11] J.N. Sørensen. Glauert revisited. In *Proceedings of EAWC conference The Science of Making Torque from the Wind*, June 2022.
- [12] J. Johansen, H. A. Madsen, M. Gaunaa, and C. Bak. Design of a wind turbine rotor for maximum aerodynamic efficiency. *WIND ENERGY*, 12:261 – 273, 2009.
- [13] J. Jonkman, S. Butterfield, W. Musial, and G. Scott. Definition of a 5-mw reference wind turbine for offshore system development. Technical Report NREL/TP-500-38060, NREL, Golden, Co, USA, 2009.
- [14] M. Korjan. Predesign rotor blade (full example). Technical Report unpublished, bewind GmbH, Rendsburg, Germany, 2021.

- [15] A. Li, M. Gaunaa, G. R. Pirrung, and S. G. Horcas. A computationally efficient engineering aerodynamic model for non-planar wind turbine rotors. *Wind Energy Science*, 7:75–104, 2022.
- [16] M. L. Buhl, Jr. A New Empirical Relationship between Thrust Coefficient and Induction Factor for the Turbulent Windmill State. NREL/TP-500-36834, NREL.
- [17] H. A. Madsen, T. J. Larsen, G. R. Pirrung, A. Li, and F. Zahle. Implementation of the blade element momentum model on a polar grid and its aeroelastic load impact. *Wind Energy Science*, 5:1–27, 2020.
- [18] NN. <https://gcc.gnu.org/fortran/>.
- [19] NN. `de.wikipedia.org/wiki/B{\unhbox\voidb@x\group\let\unhbox\voidb@x\setbox\@tempboxa\hbox{e\global\mathchardef\accent@spacefactor\spacefactor}\let\begingroup\let\typeout\protect\begingroup\def\MessageBreak{\Omega(Font)}\let\protect\immediate\write\m@ne{LaTeXFontInfo:\def{oninputline102.}\endgroup\endgroup\relax\let\ignorespaces\relax\accent18e\egroup\spacefactor\accent@spacefactor}zierkurve.`
- [20] NN. www.gnuplot.info.
- [21] O. de Vries. Fluid Dynamic Aspects of Wind Energy Conversion. Technical report, AGARDograph Bo. 242, 1979.
- [22] W. Press, A. Teukolsky, W. Vetterling, and B. Flannery. *NUMERICAL RECIPES in Fortran, 2nd Ed. The Art of Scientific Computing*. Cambridge University Press, Cambridge, UK, 1992.
- [23] Ramdin, S.F. Prandtl tip loss factor assessed. Master’s thesis, Delft University of Technology, 2017.
- [24] R.E. Wilson and P.B.S Lissaman and S.N. Walker. Aerodynamic Performance of Wind Turbines. Technical report, Oregon State University, 1976.
- [25] S.A. Ning. A simple solution method for the blade element momentum equations with guaranteed convergence. *Wind Energy*, (17):1327–1345, 2014.
- [26] Schaffarczyk, A.P. *Introduction to Wind Turbine Aerodynamics, 3rd Ed.* SpringerNature, Berlin, 2024.
- [27] S. Schmitz. *Aerodynamics of Wind Turbines*. Wiley, 2020.
- [28] Scott, Samuel and Greaves, Peter and Macquart, Terence and Pirrera, Alberto. Comparison of blade optimization strategies for the IEA 15 MW reference wind turbine. In *Proceedings of EAWC conference The Science of Making Torque from the Wind*, June 2022.

- [29] J. N. Sørensen. *General Momentum Theory for Horizontal Axis Wind Turbines*. Springer International Publishing Switzerland, 2016.
- [30] Syed, S. Aerodynamic Design of a 20MW blade. Master's thesis, Kiel University of Applied Sciences, 2022.

Draft